

Lightweight Computation Single-Image Fog Removal Based on a New Improved Adaptive Dark Channel Prior

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Abstract—This paper addresses the challenge of image quality degradation in intelligent vehicle systems caused by adverse weather conditions, particularly fog, on embedded platforms. It proposes an optimized dark channel prior (DCP)-based fog removal algorithm, optimized for real-time processing on resource-constrained platforms, such as those equipped with a single neural processing unit (NPU). The algorithm introduces an adaptive mechanism to address overexposure issues commonly found in regions with sky or sun, ensuring robust performance across diverse hazy conditions and exposure levels. Tailored for intelligent vehicle applications, the method supports real-time processing of high-definition video, delivering a speed 151.14 times faster than state-of-the-art (SoTA) learning-based approaches like FFA-Net and DehazeNet, while achieving a structural similarity index (SSIM) of up to 0.914 compared to reference fog-free images. Moreover, it significantly enhances critical vision-based tasks, such as YOLOv4 object detection, improving object detection accuracy by up to 18%. This lightweight and efficient solution makes it particularly suitable for deployment in advanced driver assistance systems (ADAS) and autonomous navigation platforms, contributing to improved safety and reliability in foggy conditions.

Index Terms—Dark channel prior, dehazing, fog removal, image enhancement, image processing.

I. INTRODUCTION

FOG, an atmospheric phenomenon caused by the scattering of light through water droplets or dust particles, significantly reduces visibility by absorbing and diffusing photons before they reach the observer. As a result, images captured

in foggy conditions often appear blurred, low in contrast, and distorted, which poses serious challenges for vision-based systems. In the context of intelligent transportation, fog removal is essential for enhancing visibility, restoring color fidelity, and improving the performance of computer vision algorithms in tasks such as object detection, classification, and tracking [1]. These functions are critical for autonomous navigation, collision avoidance, and advanced driver assistance systems (ADAS). Despite its importance, fog removal remains a challenging task, especially when dealing with single foggy images and unknown scene depth [2]. Various approaches have been proposed, including traditional image enhancement [3], [4], prior-based methods [5], and learning-based techniques. While conventional methods aim to enhance contrast and visual clarity, they can sometimes produce visually unnatural results. Among prior-based approaches, the Dark Channel Prior (DCP) has gained popularity for its effectiveness in single-image dehazing [5], and numerous enhancements have been developed to minimize artifacts and further improve output quality [6], [7], [8], [9], [10]. These advancements contribute significantly to better scene visibility, facilitating real-time tasks such as lane detection and object recognition in complex driving environments.

Autonomous vehicles that rely on perception-based sensors face significant challenges under low-visibility conditions such as fog [11]. Clear visual input is essential for perception tasks like detection, segmentation, and tracking, but fog can degrade image quality, reduce contrast, and obscure critical scene details [12]. This affects not only single images but also entire video streams, making real-time applications like surveillance and navigation unreliable. Additional issues such as motion blur or improper processing can further distort important features [13]. These factors increase the risk of incorrect predictions and unsafe decisions. As a result, there is a pressing need for fast, resource-efficient image restoration methods that preserve contextual information while supporting real-time performance. Such techniques are key to enhancing the safety and reliability of intelligent transportation systems in adverse conditions including fog, low lighting, and glare [14], [15], [16].

In addition to DCP, other priors and assumptions have been explored. For example, the color-lines prior [17] leverages the line structure in the color space to estimate transmission, while the non-local prior [18] assumes that a small number

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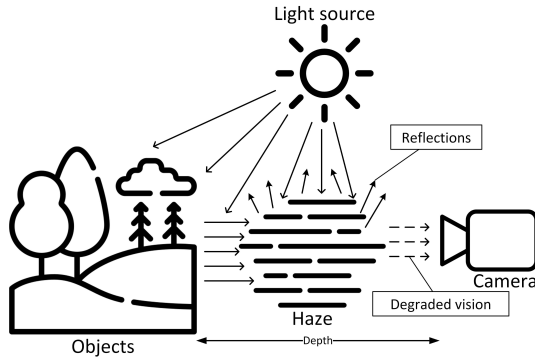


Fig. 1. Physical Scattering Model.

of distinct, clustered colors in RGB space can represent the colors of a clear image. Similarly, the color ellipsoid prior [19] assumes that the color distribution in clear regions follows an ellipsoidal shape, improving the accuracy of dehazing. These techniques contribute to enhanced fog removal and improved visibility in a variety of environments, directly benefiting the operational efficiency of intelligent vehicle systems.

Learning-based approaches have also been utilized to estimate airlight and transmission maps from foggy images using convolutional networks [20]. For example, DehazeNet [21] is a robust CNN architecture designed to estimate transmission maps from hazy images, producing clear and visually enhanced results. The Feature Fusion Attention Network (FFA-Net) [22] introduces a novel Feature Attention (FA) module that integrates pixel-level and channel-level attention to efficiently manage various image aspects. Its FFA structure further enhances dehazing by aggregating features from multiple layers while preserving low-level details. DehazeFormer [23] builds on the Swin Transformer [24], incorporating modified activations and aggregation mechanisms to achieve state-of-the-art dehazing performance, even in challenging scenarios. These learning-based methods have found significant applications in intelligent vehicles, where they enable robust and adaptive fog removal, enhancing safety and functionality under adverse weather conditions. However, they typically require large training datasets and may struggle with distribution shifts in test data or extreme haze conditions. Some models, such as [25] and [26], contain a large number of parameters, which can negatively impact computational efficiency, making them less suitable for embedded systems in autonomous vehicles.

This paper focuses on lightweight prior-based techniques, which have shown promise in real-world scenarios, especially in outdoor fog removal critical for intelligent vehicle systems. Indoor vision systems typically do not suffer from fog-related degradation, making the defogging of outdoor scenery the primary focus of this article. Fig. 1 illustrates the effect of light scattering from airborne particulate matter on visibility. The proposed method employs an anisotropic diffusion technique to refine the transmission map more effectively, significantly reducing computation time. By constructing an adaptive fog factor, our method works better in sky and

overexposed scenes, which are the limitations of previous works. Especially for embedded platforms with single NPUs used for object detection, this approach aligns with the computational constraints of intelligent vehicles. As a result, it can restore a fog-free scene with a structural similarity of up to 0.914, compared to a reference fog-free image, already reaching modern learning-based methods between FFA-Net and DehazeNet. Experimental results show that the proposed method can process 151.14 times faster than learning-based methods. At the same time, vision quality is not only more natural to the original fog-free dataset but also maintains a low computational complexity feature. When combined with object detection, such as the YOLOv4 detector, after fog-free processing, the performance can improve by up to 5-7% in object numbers and 18% detection confidence, demonstrating its practicality for intelligent vehicle systems.

The main contributions of this paper can be summarized as follows: 1) Proposing an adaptive fog removal method that optimizes ω for improved clarity of road features and obstacles, with a focus on real-time processing speeds for ADAS and autonomous navigation applications. 2) Integrating efficient contrast enhancement techniques to balance computational efficiency and image quality, ensuring fast, effective defogging for ITS. 3) Evaluating the method against state-of-the-art techniques, demonstrating superior performance in both visual quality and processing speed, suitable for real-time autonomous vehicle systems.

II. DARK CHANNEL PRIOR

The atmospheric scattering model [27], [28] is essential to understand the physics behind image defogging algorithms, as it explains how light interacts with atmospheric particles, leading to scattering and attenuation that reduce image clarity under foggy conditions. The model is mathematically expressed as:

$$I(x) = t(x)J(x) + (1 - t(x))A, \quad (1)$$

where $I(x)$ represents a foggy image, $J(x)$ is a clear image without fog, A is the global atmospheric light (or air light for short), and $t(x)$ is the transmission map describing the portion of the light that is not scattered and reaches the camera. After determining the values of A and $t(x)$, the clear image $J(x)$ can be restored from the foggy image $I(x)$ as:

$$J(x) = \frac{I(x) - A}{t(x)} + A. \quad (2)$$

The properties of air particles distinguish fog, haze, and smoke. Throughout the following sections, both “fog” and “haze” terms will be used to refer to most scenarios. This model is especially critical for autonomous navigation systems, where clear visual perception is essential for safe navigation, object detection, and real-time decision-making in autonomous vehicles and smart traffic monitoring systems.

The dark channel prior (DCP) method [5], though a well-established and widely used approach, remains effective in addressing haze and fog by leveraging its ability to estimate atmospheric light and transmission. It is a statistical analysis of haze-free outdoor images to reveal that local patches in

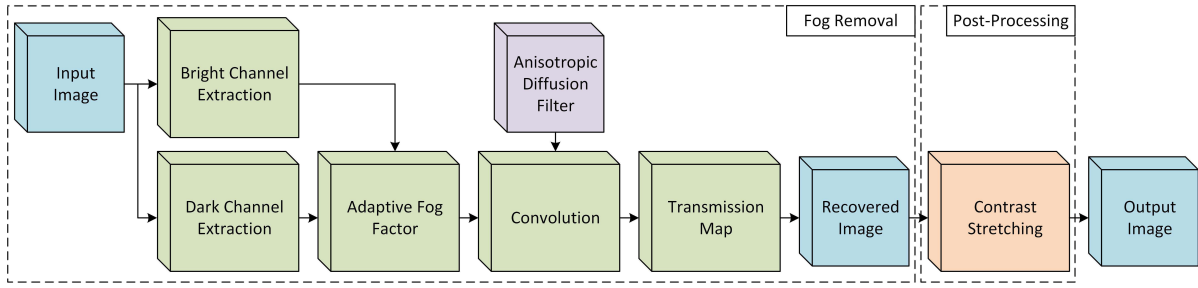


Fig. 2. Flowchart of the proposed lightweight computation fog removal method.

such images contain at least one color channel with very low brightness. The dark channel is defined as follows:

$$I^{dark}(x) = \min_{y \in \Omega(x)} \left(\min_c (I^c(y)) \right), \quad (3)$$

where c is a color channel of RGB color space, I^c represents a channel from the input image I , $\Omega(x)$ is a local patch centered at the position x . In most outdoor scenes without the sky, I^{dark} intensity is low, usually approaching zero:

$$I^{dark}(x) \approx 0. \quad (4)$$

Exploiting this assumption, the dark channel is sorted by its pixels' intensity value:

$$I^{sorted} = \operatorname{argmax}_x (I^{dark}(x)), \quad (5)$$

in which argmax returns the indices of values in the dark channel that are sorted in descending order; the indices are saved in I^{sorted} . Then, the global atmospheric light A is calculated as follows:

$$A^c = \sum_{i=0}^{N-1} I^c(I^{sorted}(i)) / N, \quad (6)$$

whereas N equals to 0.01% to 0.1% of the pixels in $I(x)$ having the highest intensity value retrieved from (5). With A^c , the DCP principle is used to calculate the dark channel of the atmospheric-scattering model:

$$\min_{y \in \Omega(x)} \left(\min_c \frac{I^c(y)}{A^c} \right) = t(x) \min_{y \in \Omega(x)} \left(\min_c \frac{J^c(y)}{A^c} \right) + 1 - t(x). \quad (7)$$

Now, by choosing a fog factor ω , the transmission map of the input scene can be estimated as follows:

$$t(x) = 1 - \omega \times \min_{y \in \Omega(x)} \left(\min_c \frac{I^c(y)}{A^c} \right). \quad (8)$$

However, to further enhance its performance, it can be refined to minimize potential side effects, such as blurring and over-smoothing, which occur when key elements of the image signal are overly compressed. These refinements offer opportunities for improving image clarity while retaining the efficiency of the DCP method.

III. PROPOSED LIGHTWEIGHT COMPUTATION METHOD

This paper proposes a computationally efficient method that integrates an enhanced dark channel prior-based fog removal algorithm with post-processing techniques to improve the visual quality of foggy images as shown in Fig. 2. The

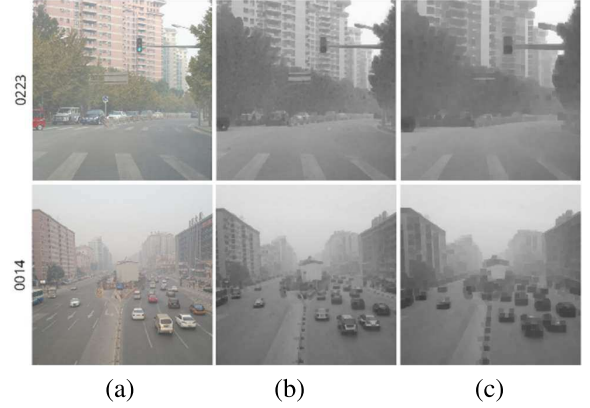


Fig. 3. Dark channel extraction: (a) Input images, (b) Dark channels extraction by 15×15 pixel Ω , (c) Dark channels extraction without Ω on RESIDE-SOTS dataset.

approach is designed to be lightweight, ensuring low time complexity, which allows for real-time processing on systems with limited computational power. By reducing the processing load while maintaining high-quality image enhancement, the method is particularly suitable for embedded platforms that require fast and efficient fog removal.

This paper proposes a computationally efficient method that integrates an enhanced dark channel prior-based fog removal algorithm with post-processing techniques to improve the visual quality of foggy images as shown in Fig. 2. The approach is designed to be lightweight, ensuring low time complexity, which allows for real-time processing on systems with limited computational power. By reducing the processing load while maintaining high-quality image enhancement, the method is particularly suitable for embedded platforms used in intelligent transportation systems, such as autonomous vehicles, smart roadside cameras, and drones for traffic monitoring, where real-time defogging under harsh environmental conditions is crucial for safety and performance.

A. Fog Removal

1) *Dark Channel Extraction*: Employing local patches leads to “blocking artifacts”, significantly when enlarging the patch size as shown in Fig. 3. To mitigate these artifacts, a necessary step is the “Transmission map refinement”, which is very time-consuming. However, the dark channel’s essence lies in the lowest intensity value among the RGB channels at

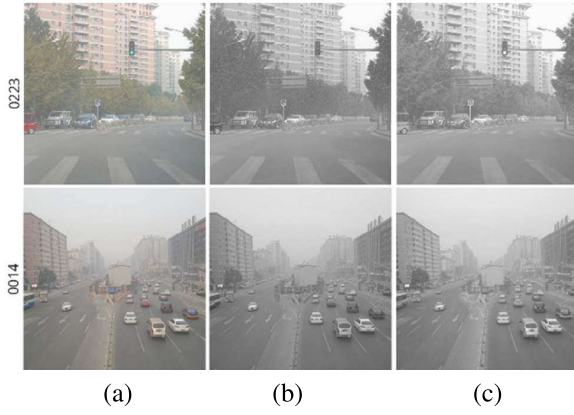


Fig. 4. Bright channel extraction: (a) Input images, (b) Dark channels, (c) Bright channels extraction without Ω on RESIDE-SOTS dataset.

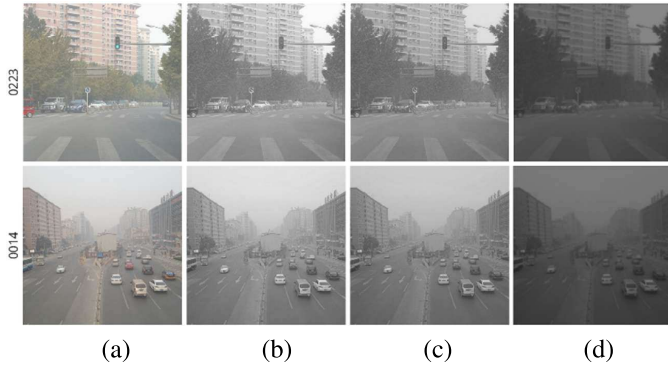


Fig. 5. Transmission Map Estimation: (a) Input images, (b) Dark channels, (c) Air light maps, (d) Transmission maps on RESIDE-SOTS dataset.

each pixel. Thus, the dark channel can be extracted without using a local patch Ω , as:

$$I^{dark}(x) = \min_c(I^c(x)). \quad (9)$$

2) *Bright Channel Extraction*: The component opposite to the dark channel is called the bright channel. The proposed method exploits the bright channel to analyze the features of the images and uses it as an additional factor to defog better, as shown in Fig. 4. The bright channel without using a local patch Ω is defined as:

$$I^{bright}(x) = \max_c(I^c(x)). \quad (10)$$

3) *Transmission Map Estimation*: In the previous works, after extracting the dark channel, the air light and transmission map are calculated as shown in (5) and (6). However, this method of calculating the air light needs to a sorting task which is time-consuming and sensitive to the image size. Therefore, a method [6] is used to estimate the air light map without sorting the dark channel's pixels.

$$A(x) = \omega I^{dark}(x), \quad (11)$$

where $A(x)$ is the air light map, and ω is the fog factor. It is essential to note that this $A(x)$ differs from A^c , which represents the global intensity of light across the entire scene.

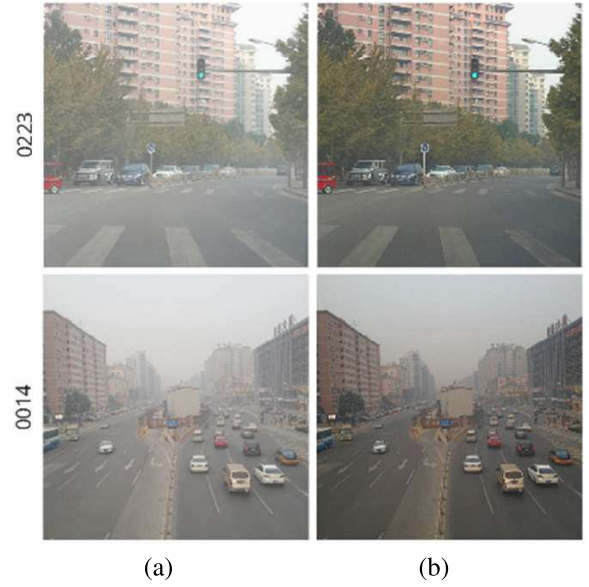


Fig. 6. Scene Recovery: (a) Input images, (b) De-fogged images on RESIDE-SOTS dataset.

Once the air light is calculated, the transmission map is estimated using anisotropic diffusion [29]:

$$t(x) = z \times \frac{\partial A}{\partial t} = z \times (\nabla c \times \nabla A + c(x, y, t) \times \Delta A), \quad (12)$$

where z is an intensity-weighted constant, Δ is the Laplacian, ∇ is the gradient, and $c(x, y, t)$ is the diffusion coefficient. Here, t represents the number of diffusion iterations. As t increases, the final image becomes progressively blurrier. The Laplacian operator forms part of the diffusion process, and the diffusion coefficient is selected as follows:

$$c(x, y) = \begin{pmatrix} 0.0625 & 0.125 & 0.0625 \\ 0.125 & 0.25 & 0.125 \\ 0.0625 & 0.125 & 0.0625 \end{pmatrix}. \quad (13)$$

To achieve a smoothing effect while preserving edge details, the values of the surrounding pixels are multiplied by 0.125, as diffused by the above operator, while the value of the central pixel is multiplied by 0.25. Fig. 5 shows the results of the modified transmission map estimation.

4) *Scene Recovery*: After retrieving the needed air light A and the transmission map $t(x)$, the fog-free scene can be recovered based on Eq. (2). Fig. 6 shows some recovered examples of fog-free images.

5) *Adaptive Fog Factor*: To optimize dehazing for vision-based transportation systems we analyze the effect of the tuning factor ω in Eq. (12). Fig. 9 shows transmission maps computed across a range of ω values (0.0–1.0, step 0.05). As illustrated in Fig. 7, increasing ω generally improves output quality linearly, enhancing visibility of road features and obstacles. This aligns with previous works [5], [6], [30], [31], [32], which often use ω between 0.9 and 0.95. However, for certain scenes (Fig. 8), overly high ω leads to quality degradation. By analyzing 500 outdoor images, we found that adjusting ω adaptively can improve results in 25.2% of cases—achieving gains up to 36.71% in PSNR and 4.49% in SSIM.

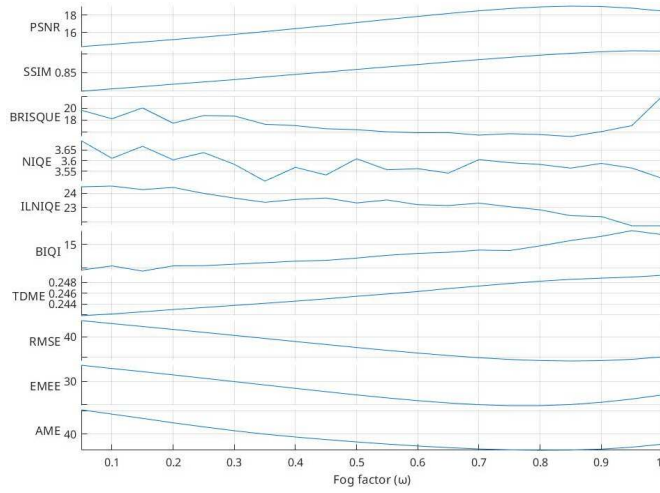


Fig. 7. The proportional increasing in quality metrics to ω on RESIDE image 0042.

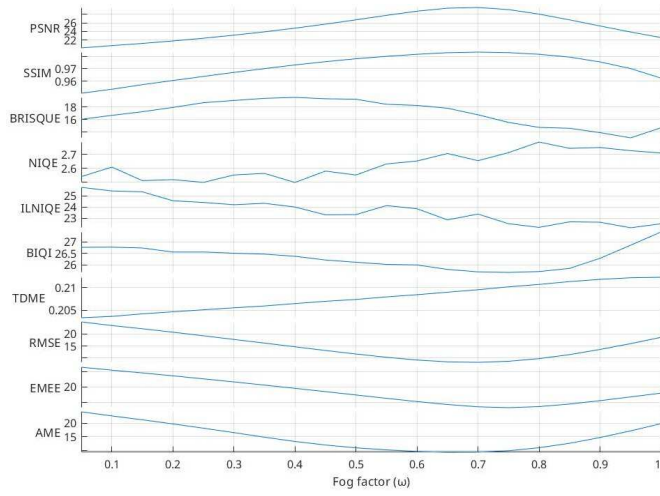


Fig. 8. Quality metrics reaching their peaks and turning to the concave downward on RESIDE image 0036.

Failure cases, such as those in Fig. 10 and Fig. 11, often stem from extreme brightness conditions or scenes that contain both dark and bright regions, common in traffic scenes with headlights, reflections, or sun glare. In such scenarios, setting $\omega = 0.6$ produces more balanced and informative outputs. To address this variability, an adaptive mechanism for selecting ω is proposed:

$$\omega = \begin{cases} 0.6, & \text{if } p > T_p, \\ 0.9, & \text{otherwise.} \end{cases} \quad (14)$$

where p is the percentage of bright channel's pixels having intensity value higher than $T_{intensity}$, where $T_p = 0.43$ is the threshold of overexposure and $T_{intensity} = 200$.

B. Post-Processing

While the adopted method [5] is computationally efficient, often results in low-contrast output, which is undesirable for vision-based driving systems. To address this, we apply a

Algorithm 1 Percentile Stretching Algorithm

- 1: Initialize p as the percentile threshold (e.g., $p = 5$);
- 2: Convert the input color image to grayscale as I_g ;
- 3: Obtain the dimensions W and H of I_g ;
- 4: Compute the histogram $\mathbb{H}$ of I_g ;
- 5: Compute the probability distribution function $\mathbb{P}$ from $\mathbb{H}$;
- 6: Calculate the cumulative distribution function (CDF) $\mathbb{C}$ from $\mathbb{P}$;
- 7: Find the index l corresponding to the lower $p\%$ in $\mathbb{C}$;
- 8: Find the index u corresponding to the upper $(100 - p)\%$ in $\mathbb{C}$;
- 9: Set the lower and upper bounds for output intensity values (o_{low} and o_{up});
- 10: Calculate thresholds t_r , t_g , and t_b for (R, G, B) color channels;
- 11: Initialize the transformation function T using t_r , t_g , t_b ;
- 12: **for** i from 1 to W **do**
- 13: **for** j from 1 to H **do**
- 14: Map pixel mask u_r , u_g , and u_b using corresponding values in T
- 15: Update out_r , out_g , and out_b with T values that masked by u_r , u_g , u_b
- 16: Combine output values of each color channel out_r , out_g , out_b to out
- 17: **end for**
- 18: **end for**

TABLE I
ORGANIZATION OF RESIDE DATASET

RESIDE (Standard)			
Subset	Number of images	Type	Environment
ITS	13990	Synthetic	Indoor
SOTS	1000	Synthetic	Both
HSTS	20	Real	Outdoor

TABLE II
ORGANIZATION OF HAZARD DATASET

HAZARD		
Number of images	Type	Environment
15	Synthetic	Outdoor

lightweight contrast enhancement technique based on percentile histogram stretching [33], [34]. As shown in Fig. 12, pixel intensities are linearly stretched between selected lower and upper percentiles (e.g., 5%–95%), applied independently to each color channel. This enhances visual clarity in both dark and bright regions, which is especially beneficial for ADAS tasks like obstacle detection in overexposed skies or underlit road areas. The algorithm is described in Algorithm 1, and Fig. 13 demonstrates improved visibility and color vibrancy, particularly in scenes with low dynamic range, which is common in real-world road environments.

C. Experimental Results

To ensure fair performance evaluation, this paper utilizes the RESIDE dataset [35], a comprehensive benchmark collection comprising foggy and fog-free images from both indoor

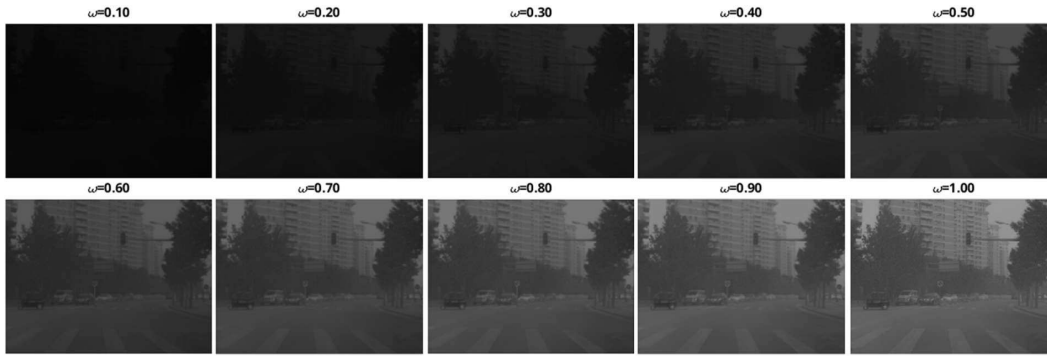


Fig. 9. Effects of choosing various ω on estimating the transmission maps.



Fig. 10. A failure sample of defogging due to overexposing on RESIDE-SOTS dataset.

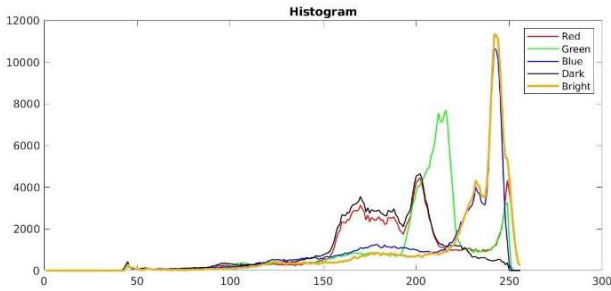


Fig. 11. Red, green, blue, dark, and bright channel histograms in a failed scene.

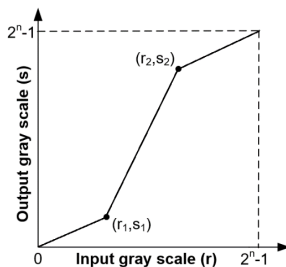


Fig. 12. Percentile histogram stretching.

and outdoor environments, specifically designed to assess single-image dehazing methods. The structure of the dataset's standard version is presented in Table I. It contains three subsets: the Indoor Training Set (ITS), the Synthetic Objective Testing Set (SOTS), and the Hybrid Subjective Testing Set (HSTS). As shown in Fig. 14, the dataset is further categorized into three groups: (a) Non-sky images, which do not contain sky regions, (b) Sky images, which include visible sky regions,

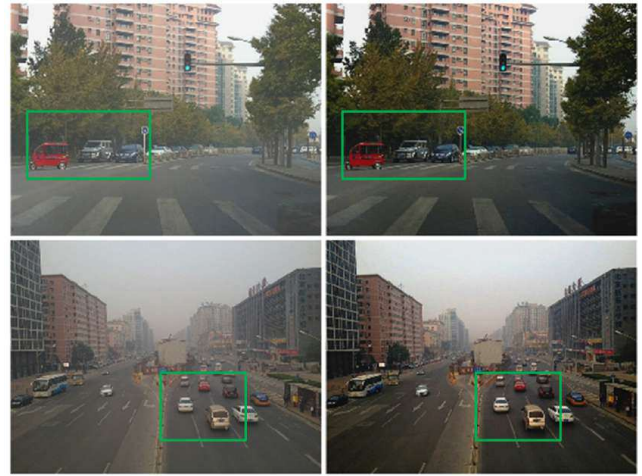


Fig. 13. Two results of contrast enhancement by this percentile stretching algorithm.

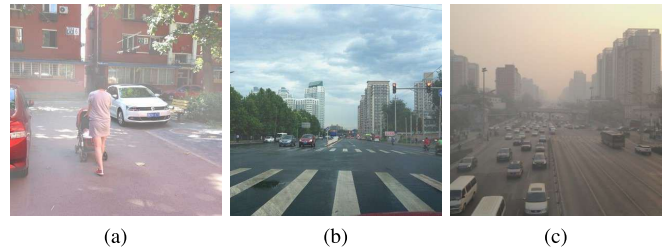


Fig. 14. Samples of comparing foggy contexts: (a) Non-sky, (b) Sky and (c) Backlight.

and (c) Backlight images, featuring strong light sources such as sunlight in the background. Additionally, we employed the HAZARD dataset [36], which consists of 15 sets of outdoor hazy images as presented in Table II, to further validate our method. Eight state-of-the-art dehazing approaches were compared in this paper: Fast Visibility Restoration (FVR) [4], Dark Channel Prior dehazing (DCP) with soft matting [5], Boundary Constrained Context Regularization (BCCR) [7], Color Attenuation Prior dehazing (CAP) [37], Non-local Image Dehazing (NLD) [18], DehazeNet [21], FFA-Net [22], and DehazeFormer [23].

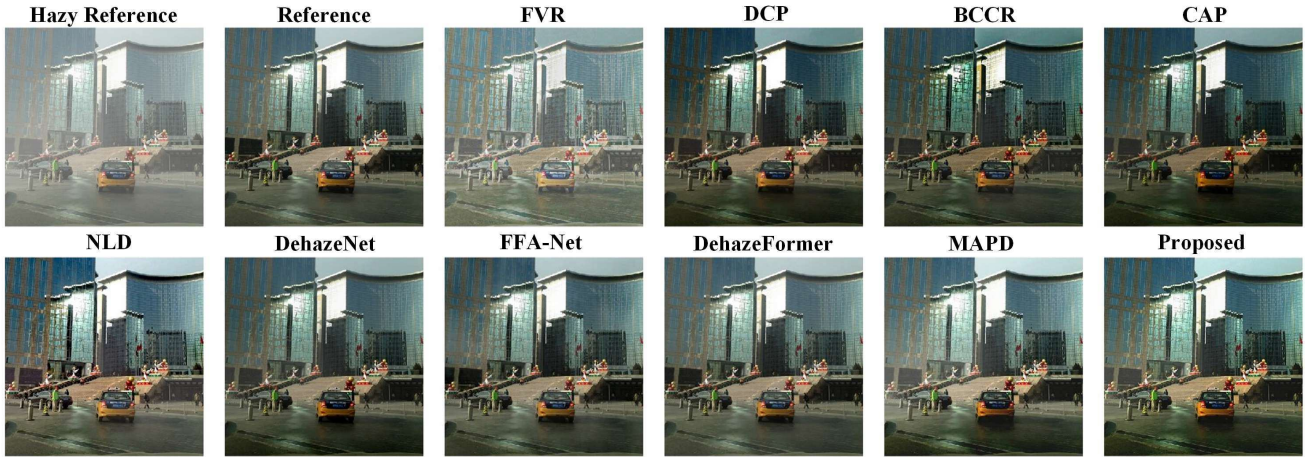


Fig. 15. Fog removal comparison on non-sky scene.

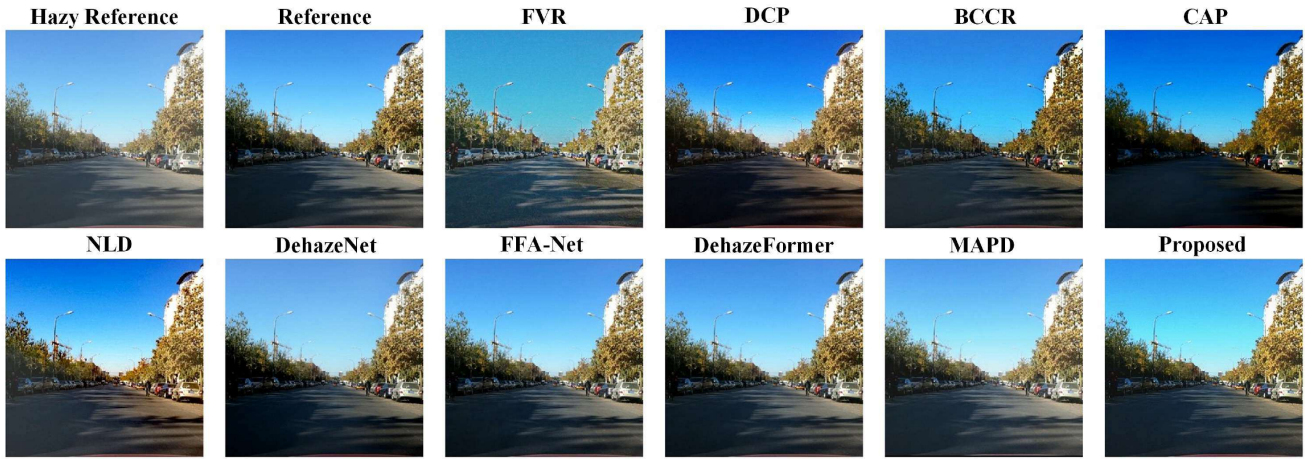


Fig. 16. Fog removal comparison on sky scene.

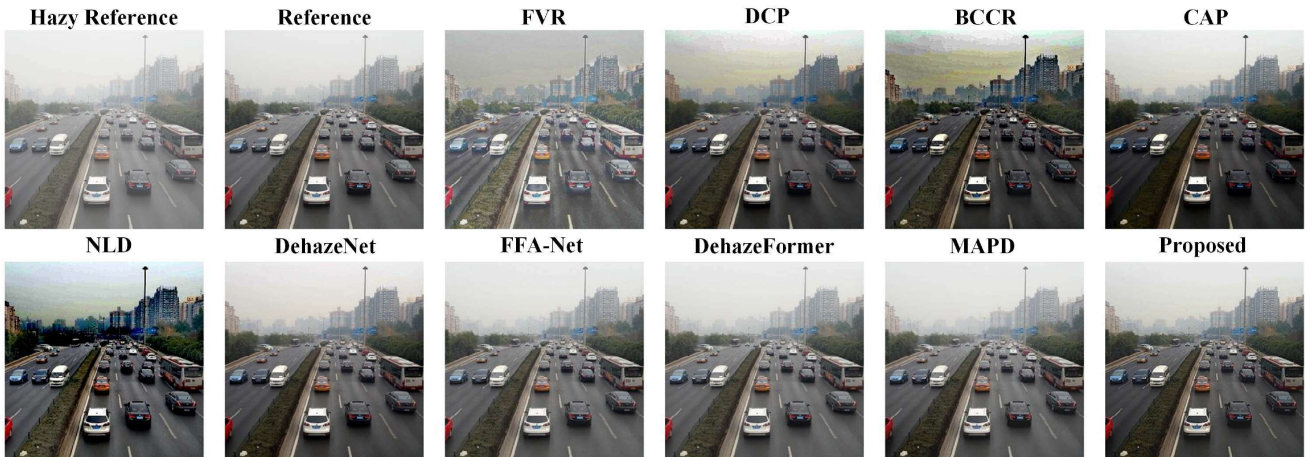


Fig. 17. Fog removal comparison on road scene.

D. Visual Perception Comparison

Figs. 15 to 17 present the visual perception comparison results for non-sky scenes, sky scenes, and backlight scenes, respectively. These figures demonstrate that while FVR [4] enhances contrast, it often leaves a thin layer of nearly overex-

posed white pixels. In contrast, methods such as DCP [5], [8], [38], BCCR [7], and the proposed approach, which utilize the dark channel prior, tend to darken the input images. However, both DCP and BCCR introduce significant color distortion in sky regions or areas with many white pixels across all figures.

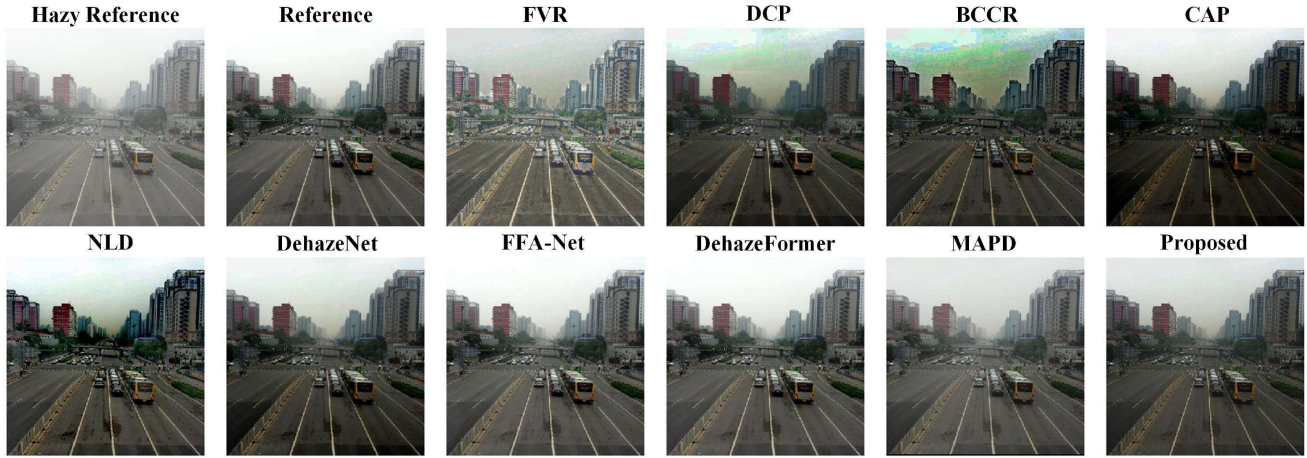


Fig. 18. Fog removal comparison on road scene.

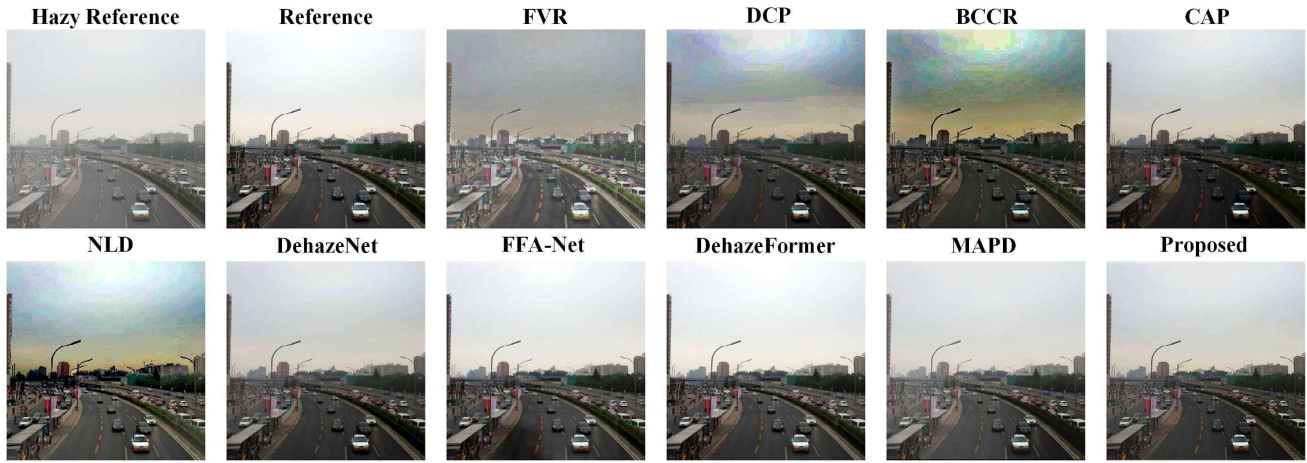


Fig. 19. Fog removal comparison on backlight condition.

TABLE III
FOG REMOVAL BENCHMARK RESULTS

Metrics	FVR 2009 [4]	DCP 2010 [5]	BCCR 2013 [7]	CAP 2015 [37]	NLD 2016 [18]	DehazeNet 2016 [21]	FFA-Net 2020 [22]	DehazeFormer 2023 [23]	MAPD 2023 [10]	Proposed
FPS	1.81	0.23	4.39	12.27	3.97	2.73	0.12	0.42	2.46	232.256
PSNR	15.4697	17.4749	14.9548	18.2211	18.2394	22.9171	30.9240	37.2181	22.0057	22.3873
SSIM	0.7809	0.8213	0.7679	0.7853	0.7839	0.8880	0.9513	0.9894	0.8809	0.9097
BRISQUE	27.0916	20.6954	17.7595	21.4680	20.8791	21.2869	23.5907	20.0990	25.2342	18.6806
NIQE	4.0945	2.6863	2.7499	2.7780	2.8156	2.6814	2.6699	2.4372	2.7570	3.0357
ILNIQE	21.2410	21.6252	20.5263	23.8528	21.3827	22.1814	21.1768	20.7208	21.7541	22.1588
BIQI	22.6782	22.2780	22.0913	21.3643	22.7991	21.5862	27.0529	21.2393	27.3156	21.5905
TDME	0.2252	0.2052	0.2195	0.2020	0.2175	0.2031	0.1911	0.2013	0.1911	0.2146
RMSE	43.5629	35.8694	45.3172	32.6485	32.2333	19.1896	7.0792	3.2406	20.9943	19.7629
EMEE	45.0364	24.6561	45.6763	37.6022	33.5287	16.8032	8.8472	3.2574	20.6766	20.2853
AME	50.8286	34.9653	47.9719	36.7598	37.4091	20.6129	12.5907	4.5148	27.0628	23.7496

Although CAP also exhibits minor color distortion in such regions, it performs better than DCP and BCCR by leveraging color attenuation to mitigate these artifacts. In addition to color distortion, DCP and BCCR face challenges in handling edge regions in backlight scenes, such as those depicted in Fig. 17. By comparison, DehazeNet, FFA-Net, DehazeFormer, and the proposed method yield the most visually consistent results, closely aligning with the reference images. Figs. 18 to 19

illustrate comparison results for scenes with reflective surfaces and road environments, respectively. Despite the diverse scene content, our proposed method consistently outperforms each state-of-the-art (SoTA) approach.

E. Quantitative Comparison

1) *Image Quality*: Table III presents the average scores of various fog removal techniques based on key quality metrics,

TABLE IV
FOG REMOVAL RUNTIME BENCHMARK RESULTS IN FPS. (*: TESTED USING CPU-ONLY PYTORCH)

Resolution	FVR 2009 [4]	DCP 2010 [5]	BCCR 2013 [7]	CAP 2015 [37]	NLD 2016 [18]	DehazeNet 2016 [21]	FFA-Net 2020 [22]	DehazeFormer 2023 [23]	MAPD 2023 [10]	Proposed
854×480	0.37	0.12	1.85	5.88	2.00	1.30	0.06	0.21	0.99	111.11
1280×720	0.08	0.05	0.89	2.63	0.79	0.49	0.03	0.09	0.50	52.63
1920×1080	0.02	0.02	0.36	1.12	0.33	0.17	0.01	0.04	0.14	15.87

including PSNR [39], SSIM [39], BRISQUE [40], NIQE [41], ILNIQE [42], BIQI [43], TDME [44], RMSE [45], EMEE [46], and AME [47]. Higher PSNR and SSIM scores, along with lower BRISQUE, NIQE, ILNIQE, BIQI, TDME, RMSE, EMEE, and AME values, indicate superior image quality. The evaluation was conducted on the SOTS subset of the RESIDE dataset and also HAZERD dataset, where the proposed method demonstrated high-quality image recovery, comparable to complex learning-based approaches such as FFA-Net [22] and DehazeNet [21]. In addition to achieving image quality on par with these methods, the proposed technique also excelled in contrast enhancement, a critical factor for objective visual perception, outperforming learning-based methods in terms of BRISQUE score.

The proposed method achieved scores of 22.3873 for PSNR, 0.9097 for SSIM, 18.6806 for BRISQUE, 3.0357 for NIQE, 22.1588 for ILNIQE, 21.5905 for BIQI, 0.2146 for TDME, 19.7629 for RMSE, 20.2853 for EMEE, and 23.7496 for AME, surpassing earlier approaches and remaining competitive with state-of-the-art learning-based methods. It is important to note that higher values indicate better quality for PSNR and SSIM, while lower values indicate better quality for BRISQUE, NIQE, ILNIQE, BIQI, TDME, RMSE, EMEE, and AME.

2) *Time Complexity*: The time complexity benchmarks on the RESIDE and HAZERD datasets are summarized in Table IV, with results measured in frames per second (FPS). Both State-of-the-Art and the proposed methods were tested using common image resolutions in digital image processing: 854×480 , 1280×720 , and 1920×1080 pixels. The tests were conducted in MATLAB R2022A on Ubuntu 20.04, using an Intel Core i5-13500 processor and 32 GB of RAM, while FFA-Net and DehazeFormer were evaluated using a CPU-only version of PyTorch in Python3. At the highest resolution of 1920×1080 pixels, the proposed algorithm demonstrates a significant performance advantage, processing images 42.89 times faster than prior-based methods and 151.14 times faster than learning-based methods, highlighting its remarkable efficiency in real-time image processing tasks. The first row of Table III also highlights the dehazing performance in terms of frames per second when processing the dataset. This substantial speed improvement is primarily due to the simplification of calculations and the unfolding of computational loops within the algorithm. By reducing the computational overhead and breaking down complex iterative operations into more straightforward, linear computations, the proposed method minimizes time-consuming processes. This streamlined approach not only accelerates the execution but also ensures efficient use of computational resources, enabling the algorithm to achieve rapid processing speeds without compromising image quality.

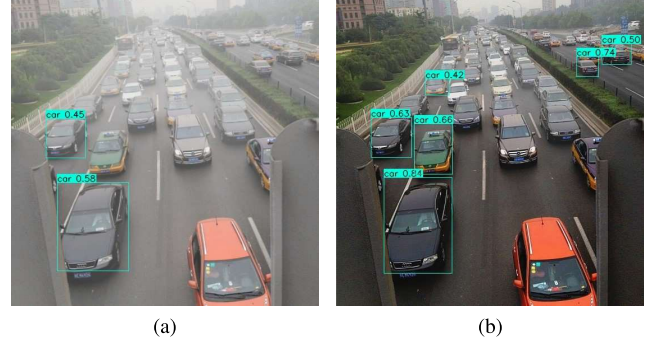


Fig. 20. Road object detection using YOLOv4 on the original image (a) and de-fogged image (b).



Fig. 21. Parked vehicle detection using YOLOv4 on the original image (a) and de-fogged image (b).



Fig. 22. Object Detection 4 using YOLOv4 on the original image (a) and de-fogged image (b).

F. Aid to Task-Driven Computer Vision

Figs. 20 to 22 demonstrate the impact of the proposed fog removal method on YOLOv4 object detection. In Fig. 20, the number of detected objects increases from 2 to 6 after de-fogging, with detection confidence improving from 0.45 to 0.63, this is very favorable especially in the foggy road conditions Fig. 21 and Fig. 22 show improved human detection in moderately foggy scenes, raising accuracy to 0.83. Fig. 23 highlights enhanced performance in backlit condi-

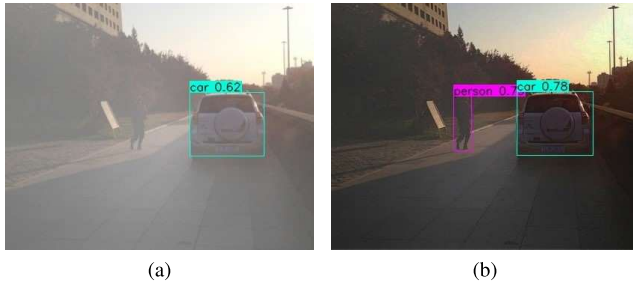


Fig. 23. Object detection on backlight condition using YOLOv4 on the original image (a) and de-fogged image (b).

tions, where the de-fogging method enables accurate detection missed in the original hazy image. This efficiency makes the method particularly well-suited for real-time applications, especially in embedded systems running YOLO object detection. By enhancing YOLOv4's ability to detect objects in foggy environments, the method improves detection rates and classification accuracy. These enhancements could achieve a 5-7% increase in object detection and up to an 18% improvement in detection confidence, solidifying this approach as a robust solution for real-time image processing in resource-constrained environments and Advanced Driver Assistance Systems (ADAS).

IV. CONCLUSION

The proposed fog removal method represents a substantial advancement over existing approaches by employing an improved dark channel prior with an adaptive fog factor, specifically designed to address challenging scenarios such as overexposed scenes and those containing both dark and bright regions. Experimental results demonstrate its superiority over current techniques, while its performance rivals that of state-of-the-art learning-based models like FFA-Net and DehazeNet. The method produces high-quality, natural-looking images, achieving an average structural similarity index (SSIM) of 0.8975, with a maximum of 0.97 when compared to fog-free images.

This approach stands out for its impressive processing efficiency, making it ideal for real-time applications, particularly in embedded systems using YOLO object detection. It significantly enhances YOLOv4's ability to identify objects in foggy conditions, improving both detection rates and accuracy. With further refinement, it has the potential to deliver even greater detection reliability, establishing itself as a robust solution for real-time image processing in resource-constrained settings.

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